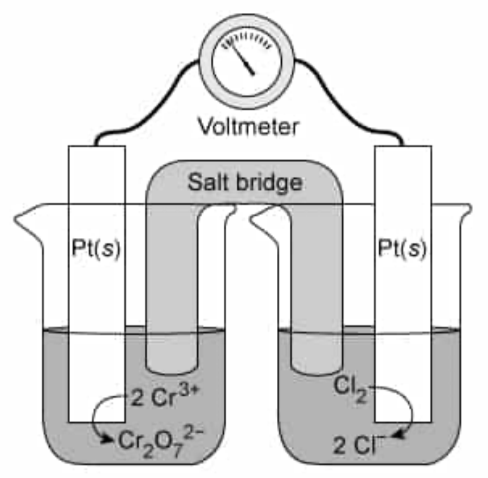
A redox reaction occurs in a galvanic cell as shown.

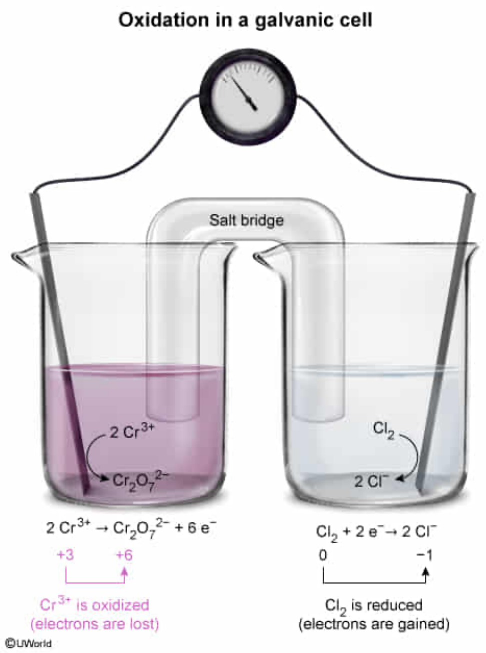


Which species is oxidized as the galvanic cell operates?

- A. Cr^{3+} [73%]
- B. Cl_2 [10%]
- C. Cl^- [5%]
- D. $Cr_2O_7^{2-}$ [11%]

Correct
72% answered correctly

Explanation:



During an oxidation-reduction (redox) reaction, electrons are **transferred** from one atom to another. The atom that **loses** electrons is **oxidized**, and the atom that **gains** electrons is **reduced**. The oxidation state of the oxidized atom **increases** by the number of electrons lost, and the oxidation state of the reduced atom **decreases** by the number of electrons gained. Oxidation states can be determined using the compound formula and oxidation state rules. The **sum** of all the **oxidation states** of each atom in the compound must be equal to the **net charge** of the compound.

In the galvanic cell shown in the question, the Cr atom has an initial oxidation state of +3 (ie, Cr^{3+}).

Applying the oxidation state rules to the $Cr_2O_7^{2-}$ anion shows that the final oxidation state of the Cr atom is +6. Although Cr is a transition state metal with a variable oxidation state, the oxidation state of O (−2) is consistent and can be used to determine the oxidation state of the Cr atoms. Because $Cr_2O_7^{2-}$ has a net charge of −2, the oxidation states of Cr and O must sum to −2.

$$Cr_2O_7^{2-} = -2$$

$$2(Cr) + 7(-2) = -2$$

$$2(Cr) - 14 = -2$$

$$Cr = +6$$

Therefore, **Cr^{3+} is oxidized** (ie, 3 electrons lost to form Cr^{6+} in $Cr_2O_7^{2-}$) as the galvanic cell operates.

(Choices B and C) The oxidation state of the Cl atom increases (ie, becomes more negative) from 0 (Cl_2) to −1 (Cl^-). As such, Cl_2 is *reduced*, not oxidized.

(Choice D) $Cr_2O_7^{2-}$ *results* from the oxidation of Cr^{3+} and is not itself oxidized.

Educational objective:
The oxidation state of each atom in a compound or ion formula can be determined using oxidation state rules. The sum of the oxidation states of each atom must equal the net charge of the compound or ion. An increased oxidation state indicates a loss of electrons (oxidation) whereas a decreased oxidation state results from gaining electrons (reduction).

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